

# Anna Fariha

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## Research Interests

My research focuses on various aspects of data system democratization: *enhancing usability* of data systems for both non-experts and experts, building *recommendation systems* for various data-management tasks, and *providing explanation frameworks* to enhance data and systems understanding.

- *Recommendation*: data summarization, data wrangling.
- *Human-Centered Data Systems & Database Usability*: query by example, data evolution, data discovery, data insights, querying unstructured documents, data transformation, student understanding, LLM-assisted data workflows.
- *Causal reasoning and explanation frameworks*: automatic program repair, data debugging, visualization debugging, explaining data non-conformance, causal insights and explanations.
- *Data Quality & Responsible ML*: data constraints, data cleaning, trusted machine learning, fair machine learning.

## Education

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| <b>Doctor of Philosophy in Computer Science</b><br><i>Manning College of Information and Computer Sciences</i> <ul style="list-style-type: none"><li>- Dissertation: Enhancing Usability and Explainability of Data Systems</li><li>- Advisor: Alexandra Meliou</li><li>- Committee members: Emery Berger, Peter Haas, Suman Nath</li><li>- CGPA: 4.00 out of 4.00</li></ul> | 2021<br><i>University of Massachusetts, Amherst, MA</i> |
| <b>Master of Science in Computer Science</b><br><i>Manning College of Information and Computer Sciences</i> <ul style="list-style-type: none"><li>- CGPA: 4.00 out of 4.00</li></ul>                                                                                                                                                                                         | 2020<br><i>University of Massachusetts, Amherst, MA</i> |
| <b>Master of Science in Computer Science and Engineering</b><br><i>Department of Computer Science and Engineering</i> <ul style="list-style-type: none"><li>- Thesis: A New Approach for Frequent Human Interaction Pattern Mining in Meeting Databases</li><li>- CGPA: 4.00 out of 4.00</li></ul>                                                                           | 2011<br><i>University of Dhaka, Bangladesh</i>          |
| <b>Bachelor of Science in Computer Science and Engineering</b><br><i>Department of Computer Science and Engineering</i> <ul style="list-style-type: none"><li>- Thesis: An Algorithm for Graph Mining using Efficient Graph Indexing for Evolving Database</li><li>- CGPA: 3.83 out of 4.00</li></ul>                                                                        | 2010<br><i>University of Dhaka, Bangladesh</i>          |

## Employment History

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|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>University of Utah</b><br><i>Assistant Professor</i><br>Kahlert School of Computing                                                              | 7/2023 – present<br>Salt Lake City, UT, USA |
| <b>Microsoft</b><br><i>Researcher</i><br>PROSE research and engineering team for AI-assisted programming<br>Managers: Gustavo Soares, Sumit Gulwani | 6/2021 – 5/2023<br>Redmond, WA, USA         |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Microsoft</b><br><i>Research Intern</i><br>PROSE research and engineering team for AI-assisted programming<br>Mentor: Ashish Tiwari<br>Project: Conformance constraints and trusted machine learning                               | 5/2019 – 11/2019<br>Bellevue, WA, USA  |
| <b>Microsoft Research</b><br><i>Research Intern</i><br>Data Management, Exploration and Mining (DMX)<br>Mentor: Suman Nath<br>Project: Adaptive interventional debugging for finding root causes of nondeterministic software bugs    | 5/2018 – 8/2018<br>Redmond, WA, USA    |
| <b>University of Massachusetts</b><br><i>Research Assistant</i><br>Data Systems Research for Exploration, Analytics, and Modeling (DREAM) Lab<br>Manning College of Information and Computer Sciences<br>Supervisor: Alexandra Meliou | 9/2016 – 5/2021<br>Amherst, MA, USA    |
| <b>Lalmatia Women's College</b><br><i>Guest Lecturer</i><br>Department of Business Administration                                                                                                                                     | 1/2015 – 4/2015<br>Dhaka, Bangladesh   |
| <b>University of Dhaka</b><br><i>Lecturer</i><br>Department of Computer Science and Engineering                                                                                                                                       | 5/2014 – 8/2016<br>Dhaka, Bangladesh   |
| <b>United International University</b><br><i>Lecturer</i><br>Department of Computer Science and Engineering                                                                                                                           | 5/2012 – 4/2014<br>Dhaka, Bangladesh   |
| <b>University of Liberal Arts</b><br><i>Competitive Programming Trainer</i><br>Coach for competitive programming in preparation for ACM ICPC Programming Contests                                                                     | 10/2011 – 1/2012<br>Dhaka, Bangladesh  |
| <b>Structured Data Systems Limited</b><br><i>Software Engineer</i><br>Android and Blackberry mobile applications                                                                                                                      | 9/2011 – 5/2012<br>Dhaka, Bangladesh   |
| <b>University of Asia Pacific</b><br><i>Competitive Programming Trainer</i><br>Coach for competitive programming in preparation for ACM ICPC Programming Contests                                                                     | 10/2010 – 10/2010<br>Dhaka, Bangladesh |

## Honors, Awards, & Achievements

|                                                                                            |      |
|--------------------------------------------------------------------------------------------|------|
| SIGMOD Distinguished PC member                                                             | 2024 |
| SIGMOD Distinguished PC member                                                             | 2023 |
| SIGMOD Distinguished PC member                                                             | 2022 |
| SIGMOD Comprehensive Reproducibility Award                                                 | 2022 |
| VLDB Best Demonstration Runner-up Award                                                    | 2020 |
| Jim Gray Scholarship, CICS, UMass Amherst                                                  | 2016 |
| Special Achievement Award: Bangladesh ICT Journalist Forum                                 | 2012 |
| Special Achievement Award: Bangladesh Bangladesh Women in IT                               | 2012 |
| University Scholarship for result in Bachelors of Science, U of Dhaka                      | 2012 |
| Special Achievement Award for outstanding performances in programming contests, U of Dhaka | 2011 |
| Dean's Honor Award (top 1%), Faculty of Engineering, U of Dhaka                            | 2011 |
| Dean's Award, Faculty of Engineering, U of Dhaka                                           | 2010 |
| Fatema-Iqbal Trust Scholarship, U of Dhaka                                                 | 2008 |

### Travel and Accommodation Scholarships

|                                                                                                                                                                |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Invited Participant with Accommodation Support, Dagstuhl Seminar on Multimodal Data Quality, Germany                                                           | 2026 |
| Invited Participant, CRA Career Mentoring Workshop, Washington, DC                                                                                             | 2024 |
| Invited Participant with Accommodation Support, Logic and Algebra for Query Evaluation<br>– Simons Institute for the Theory of Computing, Berkeley, California | 2023 |
| Invited Participant with Travel & Accommodation Support, CRA-WP Career Mentoring Workshop<br>– Chicago, Illinois                                               | 2023 |
| CRA-W Grad Cohort Workshop Travel and Accommodation Award, Washington DC                                                                                       | 2017 |

### Programming Contests

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|----------------------------------------------------------------------------------------------|------|
| Rank 5, ACM International Collegiate Programming Contest, Dhaka (Team: DU Resonance)         | 2011 |
| Rank 1, Islamic University of Technology ICT Fest Programming Contest (Team: DU Resonance)   | 2011 |
| Rank 1, Dhaka University National Collegiate Programming Contest (Team: DU Resonance)        | 2011 |
| Rank 1, Dhaka University National Collegiate Programming Contest [Women] (Team: DU Felicity) | 2011 |
| Rank 2, SUST National Collegiate Programming Contest (Team: DU Resonance)                    | 2010 |
| Rank 6, ACM International Collegiate Programming Contest, Amritapuri (Team: DU Resonance)    | 2010 |
| Rank 4, ACM International Collegiate Programming Contest, Dhaka (Team: DU Resonance)         | 2010 |
| Rank 3, ACM International Collegiate Programming Contest, Dhaka (Team: DU Ouranos)           | 2009 |
| Rank 515, Google Code Jam, (Individual, id: chorui12)                                        | 2009 |
| Rank 4, AUST National Collegiate Programming Contest (Team: DU Resonance)                    | 2009 |

## Grants

1. **Stena Center for Financial Technology – Intelligent Data Summary Recommendations for Fintech Data Analysis** [PI]. Total award amount: \$12,500. Award duration: Academic year: 2025–2026.
2. **One Utah Data Science Hub Seed Grant Award: Explaining Data Evolution** [PI]. Total award amount: \$50,000. Award duration: Mar 2024–Feb 2025.
3. **CIRC: ENS/Grand. NSF CNS 2346555: POWDER-ENS - Enhancing and Sustaining the POWDER Platform** [Co-PI, PI: Jacobus Van Der Merwe, other co-PIs: Aditya Bhaskara, Eric Eide]. Total award amount: \$4,999,980. Award duration: Oct 2024–Sep 2028.
4. **CRA-WP Distributed Research Experiences for Undergraduates (DREU): Summer support for two students.** Total award amount: \$14,000. Award duration: Summer 2024.
5. **USHE: Computer Science Targeted Workforce Application: Training the Next Generation of Database Engineers: Responsible, User-Centric, and Efficient Data Management** [led by Jeff Phillips]. Teaching grant for developing a new course: *Human-centered Data Management*. Academic year: 2023–2024.
6. **Microsoft Research Dissertation Grant.** Total award amount: \$25,000. Academic year 2020–2021.

## Publications

### Journal and Conference Publications

- [1] Anna Fariha, Lucy Cousins, Narges Mahyar, and Alexandra Meliou. Example-Driven User Intent Discovery: Empowering Users to Cross the SQL Barrier Through Query by Example. *Journal of Information Science: Special Issue on Human-In-the-Loop Data Analytics*, 2026
- [2] Tal Blau, Brit Youngmann, Anna Fariha, and Yuval Moskovitch. Causal Explanations for Disparate Trends: Where and Why? *SIGMOD*, 2026
- [3] Whanhee Cho and Anna Fariha. Data-Semantics-Aware Recommendation of Diverse Pivot Tables. *SIGMOD*, 2026
- [4] Zifan Liu, Shaleen Deep, Anna Fariha, Fotis Psallidas, Ashish Tiwari, and Avrilia Floratou. Rapidash: Efficient Detection of Constraint Violations. *Proc. VLDB Endow.*, 17(8):2009–2021, 2024

- [5] Anjali Singh, Anna Fariha, Christopher Brooks, Gustavo Soares, Austin Z. Henley, Ashish Tiwari, Chethan M, Heeryung Choi, and Sumit Gulwani. Investigating Student Mistakes in Introductory Data Science Programming. In *SIGCSE*, pages 1258–1264. ACM, 2024
- [6] Rohan Bavishi, Harshit Joshi, José Cambronero, Anna Fariha, Sumit Gulwani, Vu Le, Ivan Radicek, and Ashish Tiwari. Neurosymbolic Repair for Low-code Formula Languages. *Proc. ACM Program. Lang.*, 6(OOPSLA2):1093–1122, 2022
- [7] Sainyam Galhotra, Anna Fariha, Raoni Lourenço, Juliana Freire, Alexandra Meliou, and Divesh Srivastava. Data-Prism: Exposing Disconnect between Data and Systems. In *SIGMOD*, pages 217–231. ACM, 2022
- [8] Maliha Tashfia Islam, Anna Fariha, Alexandra Meliou, and Babak Salimi. Through the Data Management Lens: Experimental Analysis and Evaluation of Fair Classification. In *SIGMOD*, pages 232–246. ACM, 2022
- [9] Anna Fariha, Ashish Tiwari, Arjun Radhakrishna, Sumit Gulwani, and Alexandra Meliou. Conformance Constraint Discovery: Measuring Trust in Data-Driven Systems. In *SIGMOD*, pages 499–512. ACM, 2021 ([Comprehensive Reproducibility Award](#))
- [10] Anna Fariha, Suman Nath, and Alexandra Meliou. Causality-Guided Adaptive Interventional Debugging. In *SIGMOD*, pages 431–446. ACM, 2020
- [11] Anna Fariha and Alexandra Meliou. Example-Driven Query Intent Discovery: Abductive Reasoning using Semantic Similarity. *Proc. VLDB Endow.*, 12(11):1262–1275, 2019
- [12] Anna Fariha, Chowdhury Farhan Ahmed, Carson K. Leung, Md. Samiullah, Suraiya Pervin, and Longbing Cao. A New Framework for Mining Frequent Interaction Patterns from Meeting Databases. *Eng. Appl. Artif. Intell.*, 45:103–118, 2015
- [13] Amit Mandal, Mehedi Hasan, Anna Fariha, and Chowdhury Farhan Ahmed. GSCS - Graph Stream Classification with Side Information. In *APWeb*, pages 389–400, 2015
- [14] Md. Samiullah, Chowdhury Farhan Ahmed, Anna Fariha, Md. Rafiqul Islam, and Nicolas Lachiche. Mining frequent correlated graphs with a new measure. *Expert Syst. Appl.*, 41(4):1847–1863, 2014
- [15] Shafaet Ashraf, Sheikh Muhammad Sarwar, Md. Abeed Hassan, Saifuddin Md. Tareeq, and Anna Fariha. An Efficient Method for Extracting Subtrees against Forest Query. In *IMCOM*, pages 98:1–98:7. ACM, 2015
- [16] Anna Fariha, Chowdhury Farhan Ahmed, Carson Kai-Sang Leung, S. M. Abdullah, and Longbing Cao. Mining Frequent Patterns from Human Interactions in Meetings Using Directed Acyclic Graphs. In *PAKDD*, pages 38–49, 2013
- [17] Md. Samiullah, Chowdhury Farhan Ahmed, Manziba Akanda Nishi, Anna Fariha, S. M. Abdullah, and Md. Rafiqul Islam. Correlation Mining in Graph Databases with a New Measure. In *APWeb*, pages 88–95, 2013
- [18] Shariful Islam, Anna Fariha, Chowdhury Farhan Ahmed, and Byeong-Soo Jeong. EGDIM: Evolving Graph Database Indexing Method. In *ICUIMC*, pages 56:1–56:10, 2012

## Short, Demonstration, and Workshop Papers

- [19] [Aritra Mazumder](#), [Whanhee Cho](#), and Anna Fariha. QuWARTS: Query Workload Aware Relational Table Synthesis from Unstructured Text. *VLDB (Demo Track)*, 2026
- [20] [Parth Desai](#), [Aritra Mazumder](#), Fuheng Zhao, and Anna Fariha. AtomSQL: Interactive Disambiguation of NL-to-SQL via User-Guided Atom-Level Alignment. *VLDB (Demo Track)*, 2026
- [21] Tal Blau, Brit Youngmann, Anna Fariha, and Yuval Moskovitch. ExDis: Causal Explanations for Disparate Trends. *SIGMOD (Demo Track)*, 2026
- [22] [Whanhee Cho](#) and Anna Fariha. SAGE: Data-Semantics-Aware Recommendation of Diverse Pivot Tables. *SIGMOD (Demo Track)*, 2026
- [23] Bhavya Chopra, Ananya Singha, Anna Fariha, Sumit Gulwani, Chris Parnin, Ashish Tiwari, and Austin Z. Henley. Challenges in Using Conversational AI for Data Science. *HILDA@SIGMOD*, 2025

- [24] Shiyi He, Alexandra Meliou, and Anna Fariha. ChARLES: Change-Aware Recovery of Latent Evolution Semantics in Relational Data. *SIGMOD*, 2025
- [25] Ankita Sharma, Jaykumar Tandel, Xuanmao Li, Lanjun Wang, Anna Fariha, Liang Zhang, Syed Arsalan Ahmed Naqvi, Irbaz Bin Riaz, Lei Cao, and Jia Zou. DataMorpher: Automatic Data Transformation based on Large Language Models. *ICDE*, 2025
- [26] Whanhee Cho and Anna Fariha. UTOPIA: Automatic Pivot Table Assistant. *Proc. VLDB Endow.*, 2024
- [27] Bhavya Chopra, Anna Fariha, Sumit Gulwani, Austin Z. Henley, Daniel Perelman, Mohammad Raza, Sherry Shi, Danny Simmons, and Ashish Tiwari. CoWrangler: Recommender System for Data-Wrangling Scripts. In *SIGMOD*. ACM, 2023
- [28] El Kindi Rezig, Anshul Bhandari, Anna Fariha, Benjamin Price, Allan Vanterpool, Andrew Bowne, Lindsey McEvoy, and Vijay Gadepally. Examples are All You Need: Iterative Data Discovery by Example in Data Lakes. In *CIDR*, 2022
- [29] El Kindi Rezig, Anshul Bhandari, Anna Fariha, Benjamin Price, Allan Vanterpool, Vijay Gadepally, and Michael Stonebraker. DICE: Data Discovery by Example. *Proc. VLDB Endow.*, 14(12):2819–2822, 2021
- [30] Nishant Yadav, Matteo Brucato, Anna Fariha, Oscar Youngquist, Julian Killingback, Alexandra Meliou, and Peter Haas. SUBSUME: A dataset for subjective summary extraction from Wikipedia documents. In *New Frontiers in Summarization@EMNLP*, pages 131–141, 2021
- [31] Anna Fariha, Ashish Tiwari, Alexandra Meliou, Arjun Radhakrishna, and Sumit Gulwani. CoCo: Interactive Exploration of Conformance Constraints for Data Understanding and Data Cleaning. In *SIGMOD*, pages 2706–2710. ACM, 2021
- [32] Anna Fariha, Ashish Tiwari, Arjun Radhakrishna, and Sumit Gulwani. ExTuNe: Explaining Tuple Non-conformance. In *SIGMOD*, pages 2741–2744. ACM, 2020
- [33] Anna Fariha, Matteo Brucato, Peter J. Haas, and Alexandra Meliou. SuDocu: Summarizing Documents by Example. *Proc. VLDB Endow.*, 13(12):2861–2864, 2020 ([Best Demonstration Runner Up Award](#))
- [34] Anna Fariha, Sheikh Muhammad Sarwar, and Alexandra Meliou. SQuID: Semantic Similarity-Aware Query Intent Discovery. In *SIGMOD*, pages 1745–1748. ACM, 2018
- [35] Quazi Marufur Rahman, Anna Fariha, Amit Mandal, Chowdhury Farhan Ahmed, and Carson K. Leung. A Sliding Window-Based Algorithm for Detecting Leaders from Social Network Action Streams. In *IEEE/WIC/ACM WI-IAT*, pages 133–136, 2015

## PhD Thesis

- [36] Anna Fariha. *Enhancing Usability and Explainability of Data Systems*. PhD thesis, University of Massachusetts Amherst. [https://scholarworks.umass.edu/dissertations\\_2/2311/](https://scholarworks.umass.edu/dissertations_2/2311/)

## Software, Educational Content, and Patents

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- SAGE: Data-Semantics-Aware Recommendation of Diverse Pivot Tables. [Link](#)
- ChARLES: Change-Aware Recovery of Latent Evolution Semantics in Relational Data. [Link](#)
- ExDis: Explaining Disparate Trends. [Link](#)
- UTOPIA: Automatic Pivot Table Assistant. [Link](#)
- SQuID: Semantic similarity-aware Query Intent Discovery. [Link](#)
- Conformance Constraints Discovery: Measuring Trust in Data-Driven Systems (Winner of the ACM SIGMOD 2022 Most Reproducible Paper Award). [Link](#)
- SubSumE Dataset for Subjective Document Summarization. [Link](#)
- A Data Management Perspective on Fair Classification: An Experimental Analysis and Evaluation. [Link](#)
- Educational video on “Research in Computer Science (Databases/Data Management Track)”. [Link](#)

- Bavishi, R. J., Cambronero Sánchez, J. P., **Fariha, A.**, Gulwani, S., Le, V. M., Radicek, I., Simmons, D. G., & Tiwari, A. (2025). Language-agnostic computer program repair engine generator (U.S. Patent No. 12,265,789). U.S. Patent and Trademark Office. [Link](#)
- Deep, S., Liu, Z., Tiwari, A., **Fariha, A.**, Floratou, A., & Pasallidas, F. (2024). System and method for fast constraint discovery on relational data (U.S. Patent Application No. 18/060,500). United States Patent and Trademark Office. [Link](#)

## Panels and Proposal Reviews

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- Panelist – Grad-CS Women Panel, KSoC, University of Utah 2025
- Reviewer – Israel Science Foundation March, 2025
- Panelist – Northwest Database Society Annual Meeting: Panel on AI and Data Management February, 2024
- Panelist – NSF Information and Intelligent Systems division (IIS) 2023
- Panelist – NSF’s CSGrad4US Panel: “What I wish I knew before I started graduate school” October, 2023
- Panelist – SIGMOD New Researcher Symposium June, 2022
- Round Table Chair – VLDB: Interactive Querying and Visualization for Large Data August, 2021

## Invited Talks

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- *Understanding Data through Change Summarization and Causal Disparity Explanations*
  - Data Science Seminar, University of Utah October, 2025
  - TDSAI Lab, Department of Computer Science, Institute of Science Tokyo May, 2025
  - Onizuka Lab, Osaka University May, 2025
  - Database Seminar, Cornell University April, 2025
- *Explaining Data Evolution*
  - DELPHI Data Science Symposium, University of Utah November, 2024
- *Recent Trends in Scalable and Human-centric Data Systems*
  - Research Challenges in Computing Seminar Series, University of Utah October, 2023
- *Through the Data Management Lens: Experimental Analysis and Evaluation of Fair Classification*
  - Bias, Ethics and Trustworthiness in Data Science and AI event, University of Utah September, 2023
- *Blame the data, not the system: how data constraints can help in trustworthy machine learning and explain causes of data-system malfunction*
  - Data Science Lecture Series, University of Utah September, 2023
  - University of Washington, Northwest Database Society November, 2021
- *Blame the data, not the system: How data profiles can help explain causes of data-system malfunction*
  - Northwest Database Society (NWDS) Annual Meeting May, 2023
- *Enhancing usability and explainability of data systems*
  - Database Seminar, University of California San Diego (virtual) April, 2023
  - Cornell University April, 2022
  - Georgia University of Technology April, 2022
  - University of California, Santa Barbara March, 2022
  - University of Minnesota (virtual) March, 2022
  - University of Waterloo (virtual) March, 2022
  - University of Utah February, 2022
  - University of Toronto (virtual) February, 2022
  - Pennsylvania State University (virtual) February, 2022
  - EPIC meeting, University of California, Berkeley (virtual) February, 2022
  - Ohio State University (virtual) January, 2022
  - CSAIL, Massachusetts Institute of Technology (virtual) October, 2021
  - Microsoft (virtual) January, 2021

- Megagon Labs (virtual)
- University of Pennsylvania (virtual)
- DATA Lab, Northeastern University (virtual)

February, 2021  
February, 2021  
January, 2021

## Teaching

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### • University of Utah

- CS 6959/3960 - Human-Centered Data Management (graduate + undergraduate) Spring 2026
- CS 6530 - Advanced Database Systems (graduate) Fall 2025
- CS 6959/3960 - Human-Centered Data Management (graduate + undergraduate) Spring 2025
- CS 5353/6353 - Deep Learning (graduate + undergraduate) Fall 2024
- CS 6959/3960 - Human-Centered Data Management (graduate + undergraduate) Spring 2024
- CS 5353/6353 - Deep Learning (graduate + undergraduate) Fall 2023

### • University of Dhaka, Bangladesh

- CSE 1201 - Fundamentals of Programming (undergraduate) Spring 2014
- CSE 1211 - Fundamentals of Programming Lab (undergraduate) Spring 2014
- CSE 1102 - Discrete Mathematics (undergraduate) Fall 2014
- CSE 2101 - Data Structures and Algorithms (undergraduate) Spring 2015
- CSE 2111 - Data Structures and Algorithms Lab (undergraduate) Spring 2015
- CSE 3113 - Microprocessor and Assembly Language Lab (undergraduate) Fall 2015
- CSE 1111 - Fundamentals of Computer and Computing Lab (undergraduate) Fall 2015
- CSE 2112 - Object Oriented Programming Lab (undergraduate) Spring 2016
- CSE 2212 - Design and Analysis of Algorithms Lab (undergraduate) Spring 2016
- CSE 4111 - Artificial Intelligence Lab (undergraduate) Spring 2016

### • United International University, Bangladesh

- CSI 219 - Discrete Mathematics (undergraduate)
- CSI 121 - Structured Programming Language (undergraduate)
- CSI 122 - Structured Programming Language Laboratory (undergraduate)
- CSI 228 - Algorithms Laboratory (undergraduate)
- CSI 233 - Theory of Computing (undergraduate)
- CSI 341 - Artificial Intelligence (undergraduate)
- CSI 342 - Artificial Intelligence Laboratory (undergraduate)

## Student Supervision

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### PhD Advisor: Current

- Whanhee Cho Fall 2023 – Present
- Anirudh Kamath Fall 2025 – Present
- Shamit Fatin Fall 2025 – Present
- Aritra Mazumder Fall 2025 – Present

### PhD Advisor: Past

- Chetan Bajaj Spring 2025
- Shiyi He Fall 2023–Fall 2024

### Thesis Committee Member

- Anastasiia Tkachenko (PhD)
- Anirudh Kamath (MS) Spring 2025
- Kutay Eken (MS) Spring 2026

### Other PhD, Masters, & Undergraduate Students

- University of Utah
  - Elisa Wang (MS) Spring 2026 – Present
  - Parth Desai (MS) Spring 2026
  - Giang Ta (MS) Spring 2026
  - Anastasiia Tkachenko (PhD) Spring 2026
  - Kuangfei Long, Boston University Summer 2024 – Spring 2026
  - Tal Blau, Technion - Israel Institute of Technology Spring 2024 – Spring 2026
  - Shreya Raj, UMass Amherst Summer 2024 – Fall 2024
  - Vijaysurya Vempati (MS) Fall 2024
  - William Erignac Summer 2024
  - Ziming Wang Fall 2024
  - Holden Ellsworth (MS) Summer 2024
  - Bhavya Chopra, University of California, Berkeley Summer 2024
  - Yuqing Wang, University of Wisconsin-Madison (DREU by CRA, Project Report [37]) Summer 2024
  - Rania Saber, University of California-Riverside (DREU by CRA, Project Report [38]) Summer 2024
- University of Dhaka
  - Quazi Marufur Rahman (MS) 2015
  - Amit Mandal 2015
  - Mehedi Hasan 2015
  - Nitish Ranjan Bhowmik 2015
  - Kazi Mazbah Uddin 2015

## Mentorship

- Microsoft
  - Bhavya Chopra (Predoctoral Research Fellow) [PhD student at UC Berkeley] 2022–2023
  - Yang Shi (Research Intern) [Assistant Professor at Utah State University] 2022
  - Chethan Mahadevaswamy (Predoctoral Research Fellow) [Data Scientist at Microsoft] 2021–2022
- University of Massachusetts Amherst
  - Maliha Islam (PhD) [Now at Microsoft] 2020–2021
  - Hasnain Heickal (PhD) 2021–2022
  - Oscar Youngquist (MS) 2021–2022
  - Julian Killingback (MS) 2021
  - Kanchi Masalia (MS) 2020
  - Lucy Cousins (MS) 2019–2021
  - Genglin Liu (Undergrad) 2020
  - Armand Asnani (Undergrad) 2019–2020
  - Nischal Dave (Undergrad) 2019
  - Zoey (Jingyi) Sun (Undergrad from Smith College) 2018
  - Lauren Beryl Larson (REU) (Undergrad from Wellesley College) 2017
  - Michael Satanovsky (High-school Intern from Hopkins School) 2017

## Professional Service

### Internal Service

- Member, Course Review Sub Committee, KSoC 2025
- Core Member, Utah Center For Data Science 2025–present
- Faculty liaison for the Graduate Student Advisory Committee (GradSAC), KSoC 2025–present
- Member, Committee to Develop BS in Artificial Intelligence, KSoC 2025
- Member, Faculty Recruiting Committee, KSoC 2025
- Graduate Admission Committee, KSoC 2025

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|----------------------------------------------------------------------------------------------------------------------------|--------------|
| • Member, Research Support/Opportunities Committee, KSoC                                                                   | 2024–present |
| • Panelist, Graduate Student Mentoring Panel on “How to get a job in industry?”, KSoC                                      | 2024         |
| • Founder and Advisor, Grad-CS Women, KSoC ( <a href="https://gradcswomen-utah.github.io">gradcswomen-utah.github.io</a> ) | 2024–2025    |
| • Graduate Admission Committee, KSoC                                                                                       | 2024         |
| • Faculty Advisor, Women in Computing ( <a href="https://wic.utahclubs.org">wic.utahclubs.org</a> )                        | 2023–2025    |
| • Mentor of 1st Year PhD Student (Ishrat Jahan Eliza), KSoC                                                                | 2023         |
| • Co-founder of UtahDB Research Center ( <a href="https://mod.cs.utah.edu">mod.cs.utah.edu</a> )                           | 2022–present |

## External Service

### Conference Service

|                                                                                                |      |
|------------------------------------------------------------------------------------------------|------|
| • Meta Reviewer, SIGMOD (demonstration track)                                                  | 2027 |
| • Program Committee, SIGMOD (research track)                                                   | 2027 |
| • Chair, Workshop on Systems for Data-centric Agents with Human-in-the-loop (DASHSys), at VLDB | 2026 |
| • Chair, Workshop on Human-In-the-Loop Data Analytics (HILDA), at SIGMOD                       | 2026 |
| • Chair, New Researcher Symposium and Mentorship, SIGMOD                                       | 2026 |
| • Chair, Publication and Proceedings, ICDE                                                     | 2026 |
| • Session Chair, “Data Models & Interfaces” at SIGMOD                                          | 2025 |
| • Program Committee, SIGMOD (research track)                                                   | 2026 |
| • Chair, Data-AI Systems (DAIS) Workshop, at ICDE                                              | 2025 |
| • Program Committee, ICDE (research track)                                                     | 2025 |
| • Program Committee, VLDB (demonstration track)                                                | 2025 |
| • Program Committee, VLDB (research track)                                                     | 2025 |
| • Program Committee, SIGMOD (demonstration track)                                              | 2025 |
| • Program Committee, SIGMOD (research track)                                                   | 2025 |
| • Program Committee, GUIDE-AI: SIGMOD Workshop                                                 | 2024 |
| • Program Committee, ICDE (demonstration track)                                                | 2024 |
| • Program Committee, SIGMOD (demonstration track)                                              | 2024 |
| • Program Committee, SIGMOD (research track)                                                   | 2024 |
| • Program Committee, ICDE (Ph.D. Symposium)                                                    | 2023 |
| • Program Committee, SIGMOD (research track)                                                   | 2023 |
| • Program Committee, VLDB (research track)                                                     | 2023 |
| • Program Committee, SIGMOD (research track)                                                   | 2022 |
| • Program Committee, VLDB (research track)                                                     | 2022 |
| • Program Committee, EDBT (demonstration track)                                                | 2022 |
| • Program Committee, SIGMOD (research track)                                                   | 2021 |
| • Session chair, EDBT                                                                          | 2021 |
| • Program Committee, EDBT (demonstration track)                                                | 2021 |
| • Program Committee, VLDB (demonstration track)                                                | 2021 |

### Other Reviewing and Community Service

|                                                                                    |           |
|------------------------------------------------------------------------------------|-----------|
| • Reviewer, Elsevier Information Systems Journal                                   | 2025      |
| • Mentor of Xuan Luo (Simon Fraser University) at New Researcher Symposium, SIGMOD | 2024      |
| • Reviewer, the TKDE Journal                                                       | 2024      |
| • Reviewer, the VLDB Journal                                                       | 2023      |
| • Reviewer, the VLDB Journal                                                       | 2021      |
| • Reviewer, UIST                                                                   | 2021      |
| • External reviewer, CHI                                                           | 2021      |
| • External reviewer, SIGMOD                                                        | 2018–2020 |

### Problem Setter & Judge of Competitive Programming Contests

|                                                                                |      |
|--------------------------------------------------------------------------------|------|
| • State University Bangladesh Inter-University Programming Contest, Bangladesh | 2015 |
| • SAARC Programming Contest, Bangladesh University, Bangladesh                 | 2014 |
| • Islamic University of Technology ICT Fest, Bangladesh                        | 2014 |

- SAARC Programming Contest, Bangladesh University, Bangladesh 2013
- Bangladesh Informatics Olympiad, National Round, Bangladesh 2013
- North South University Inter-University Programming Contest, Bangladesh 2013
- ACM International Collegiate Programming Contest, Dhaka, Bangladesh 2012